OSTİM TEKNİK ÜNİVERSİTESİ
ANKARA

Acadia

Academic Productivity & Student Wellbeing Management System



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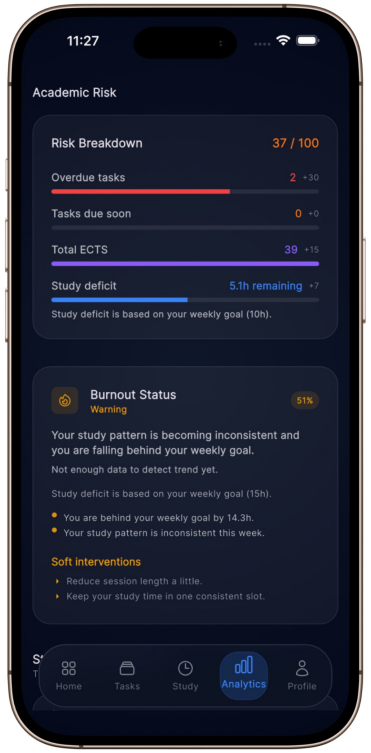
Why Acadia?

University students face increasing academic pressure every semester.
Students struggle with:

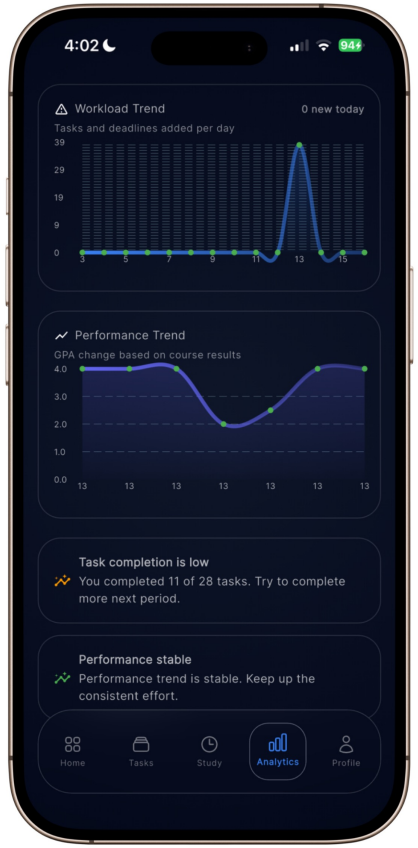
- Missed deadlines
- Poor study organization
- Inconsistent study habits
- Burnout and stress
- Difficulty prioritizing academic work
- Most productivity applications only manage tasks.
- They do not understand academic behavior.
- Acadia was designed to solve this problem.

Acadia combines:

- Academic management
- Analytics
- Realtime monitoring
- Intelligent recommendations
- AI assistance
- Wellbeing tracking
- Inside one unified system.







Problem Statement

Students often use multiple disconnected applications for:

- Tasks
- Calendars
- Reminders
- Notes
- Timers
- Analytics

This creates:

- Fragmented workflows
- Poor academic visibility
- Weak deadline management
- Inconsistent study tracking

Current systems fail to:

- Evaluate academic risk
- Monitor burnout
- Generate intelligent recommendations
- Provide realtime analytics

Acadia addresses these limitations through an integrated academic productivity ecosystem.

Project Objectives

The main objectives of Acadia are:

1. Improve academic organization
2. Monitor productivity
3. Reduce academic stress
4. Improve workload management
5. Provide intelligent recommendations
6. Support academic decision making
7. Deliver realtime insights

Acadia also provides a scalable architecture for future AI-driven academic systems.

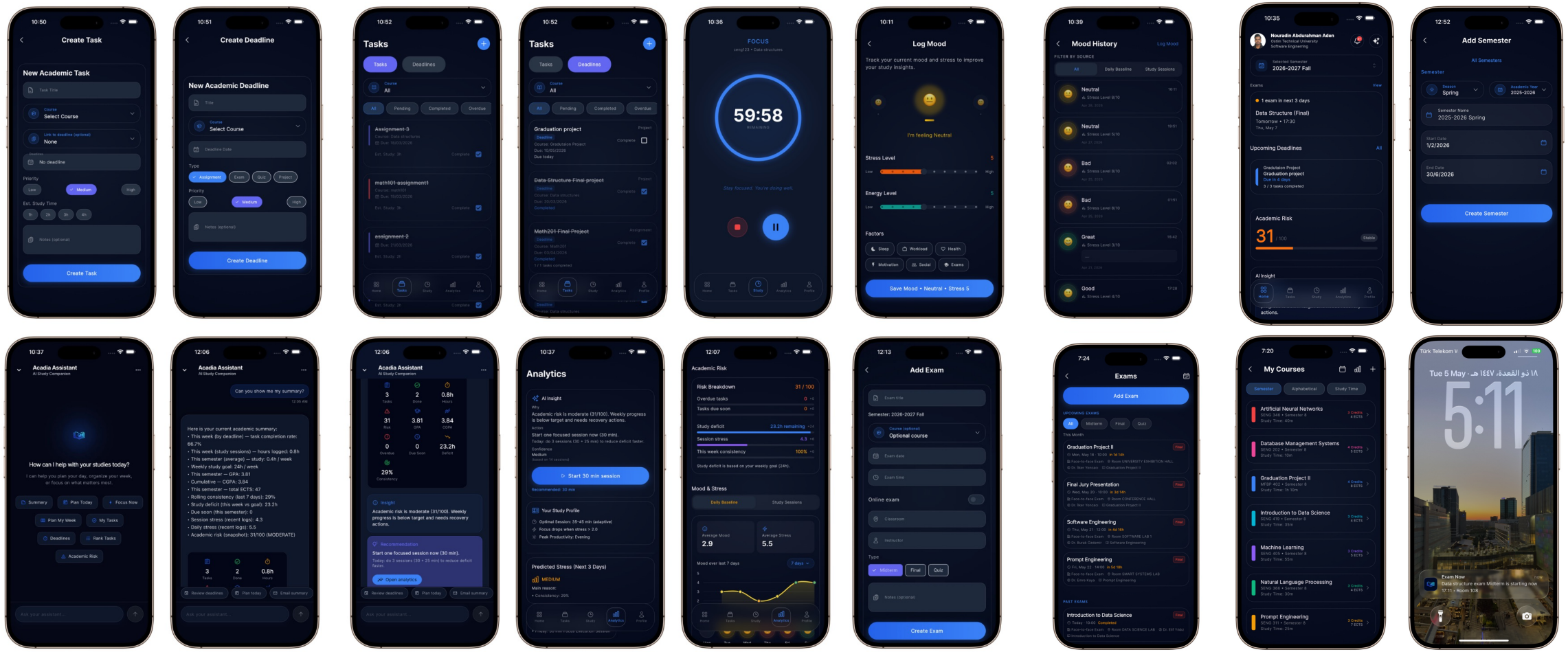
Proposed Solution

Acadia is a cross-platform academic productivity system developed using Flutter and Supabase.

Acadia provides:

1. Semester management
2. Course management
3. Task tracking
4. Deadline management
5. Study sessions
6. Pomodoro system
7. Mood tracking
8. Analytics dashboards
9. Academic risk monitoring
10. AI recommendations
11. Intelligent notifications
12. Realtime synchronization

Proposed Solution



Technology Stack

Layer	Technologies
Frontend	Flutter, Dart
State Management	Riverpod
Backend	Supabase
Database	PostgreSQL
Notifications	Firebase Cloud Messaging
AI Integration	NVIDIA NIM, Llama 3.1
Local Storage	Hive
Navigation	GoRouter
Charts	fl_chart
Version Control	GitHub
Project Management	Notion Scrum Workspace

Frontend Architecture

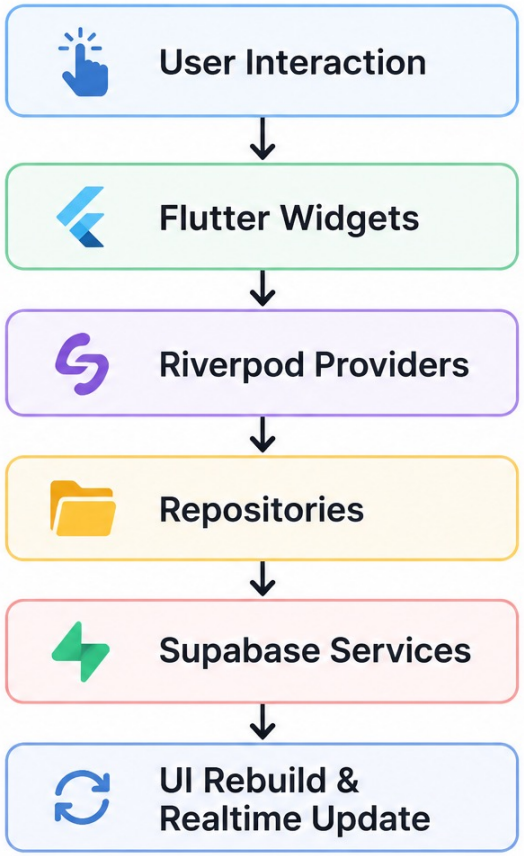
Acadia's frontend was developed using Flutter with a modular UI architecture and Riverpod state management.

The frontend system was divided into:

- Reusable widgets
- Feature-based screens
- Providers
- Controllers
- Responsive layouts
- Shared UI components

The application UI was designed to support:

- Android
- iOS
- Tablets
- Large screens
- Responsive resizing



UI/UX Design & User Experience

The UI/UX system was designed to improve:

- Readability
- Simple navigation
- Academic organization
- Student focus
- Productivity tracking

The interface follows:

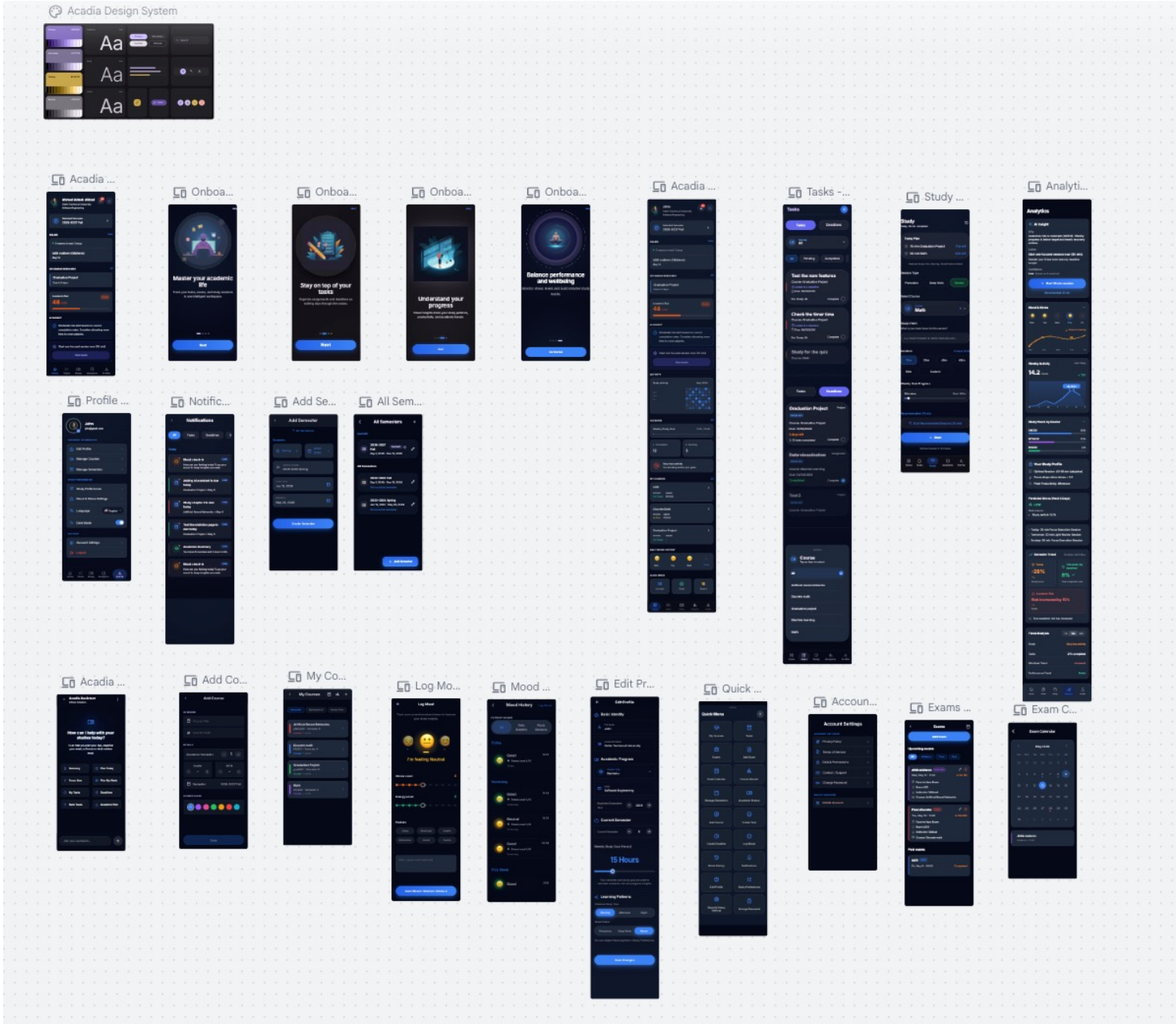
- Modern mobile design
- Glassmorphism UI
- Consistent spacing
- Clear visual hierarchy
- Responsive interactions

Design Goals

- Reduce cognitive overload
- Improve task visibility
- Simplify academic workflows
- Reduce navigation complexity
- Support long term usability

UI Features

- Animated onboarding
- Dashboard cards
- Analytics charts
- Progress indicators
- Bottom navigation
- Realtime updates
- Adaptive widgets
- Mood tracking
- Risk indicators
- Study timers



Software Development Process

Agile Scrum Development

The development process followed Agile Scrum methodology. The project was divided into weekly sprints.

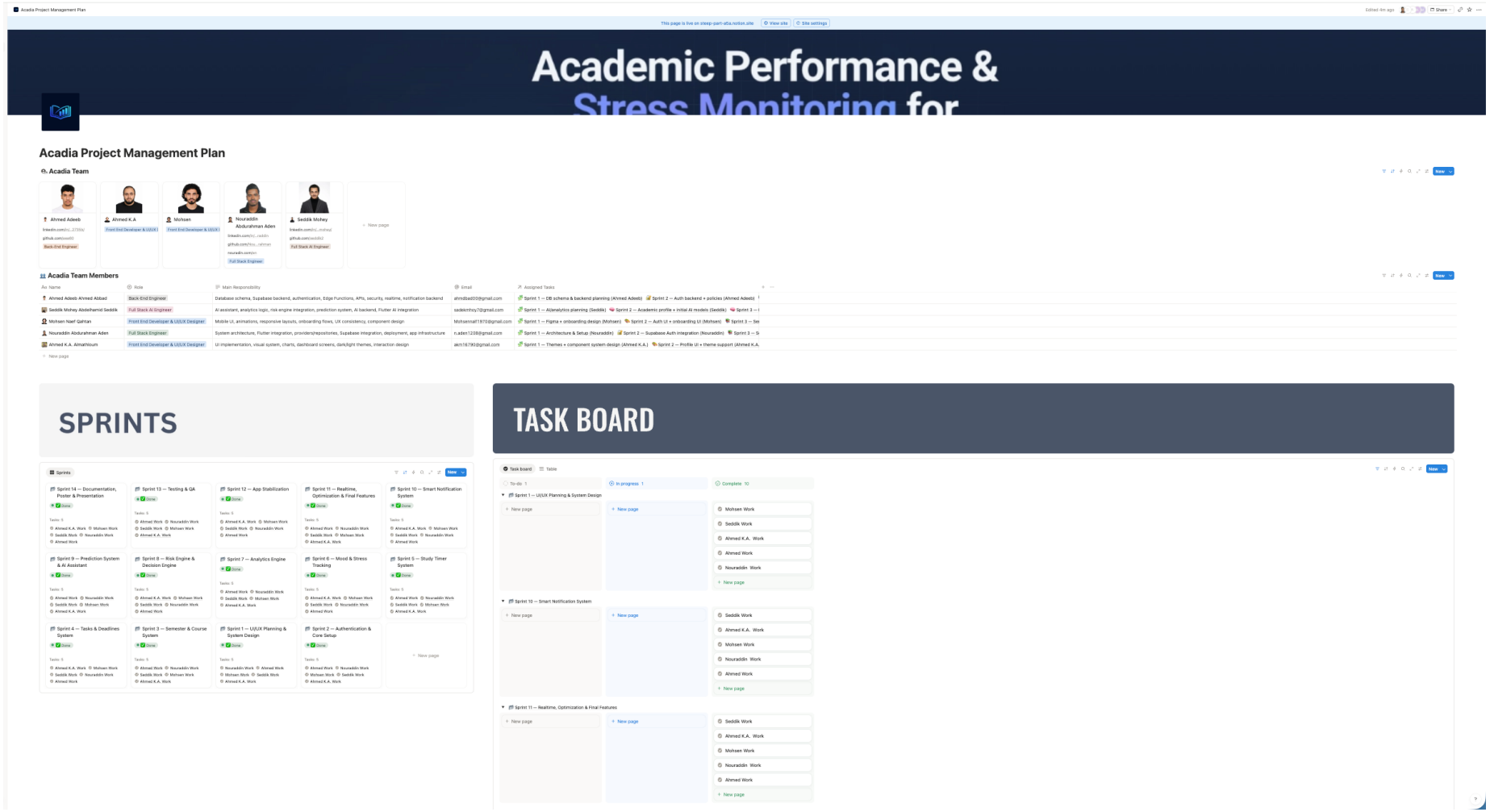
Scrum was used for:

- Sprint planning
- Backlog management
- Feature tracking
- Testing coordination
- Sprint reviews
- Retrospectives

Sprint	Main Focus
Sprint 1	UI/UX Planning & System Design
Sprint 2	Authentication & Core Setup
Sprint 3	Semester & Course System
Sprint 4	Tasks & Deadlines System
Sprint 5	Study Timer System
Sprint 6	Mood & Stress Tracking
Sprint 7	Analytics Engine
Sprint 8	Risk Engine & Decision Engine
Sprint 9	Prediction System & AI Assistant
Sprint 10	Smart Notification System
Sprint 11	Realtime Optimization & Final Features
Sprint 12	App Stabilization
Sprint 13	Testing & Quality Assurance
Sprint 14	Documentation, Poster & Presentation

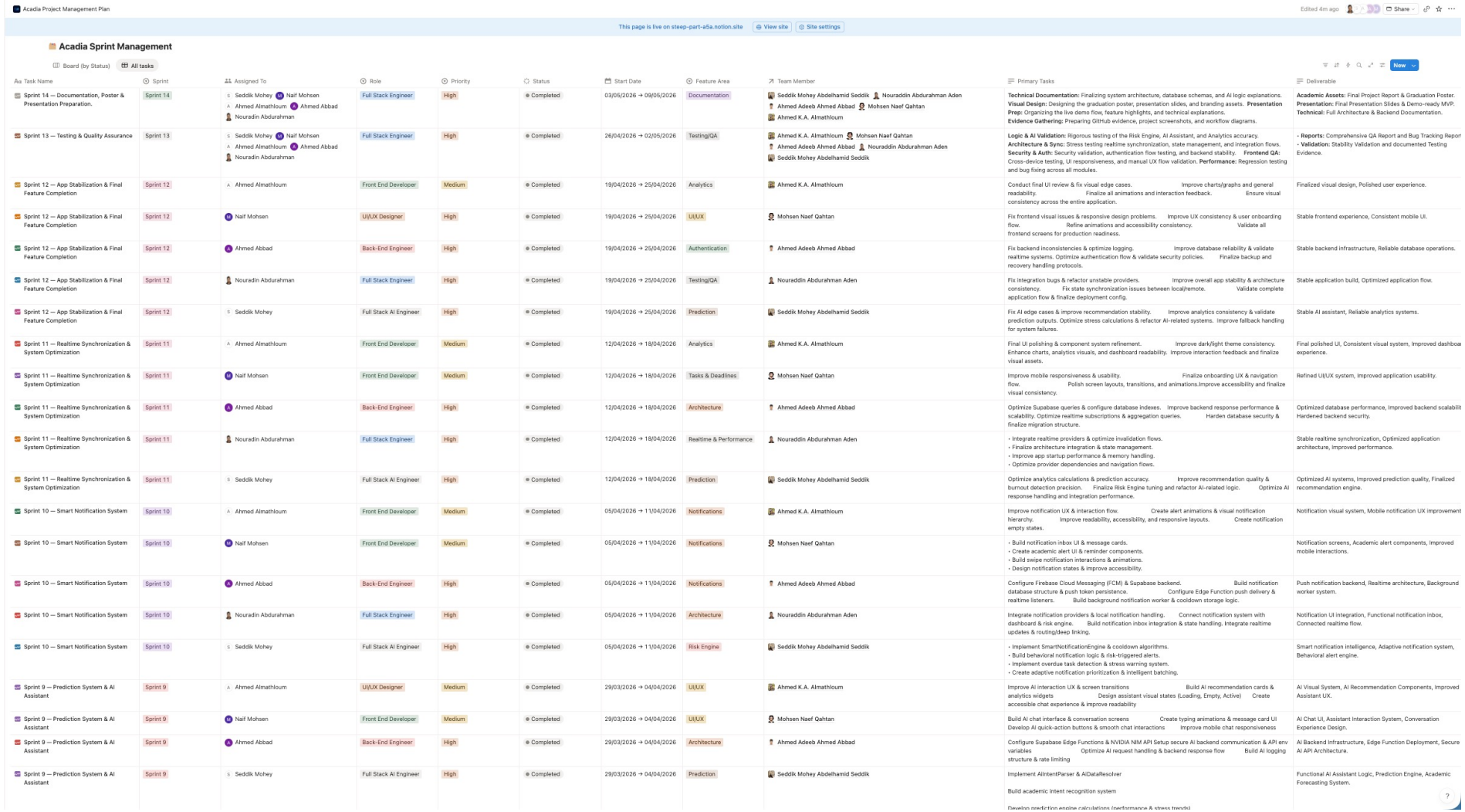
Software Development Process

Agile Scrum Development



Software Development Process

Agile Scrum Development



Acadia Project Management Plan

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Acadia Sprint Management

Board (by Status) | All tasks

Task Name	Sprint	Assigned To	Role	Priority	Status	Start Date	Feature Area	Team Member	Primary Tasks	Deliverable
Sprint 14 — Documentation, Poster & Presentation Preparation.	Sprint 14	Seddik Mohey, Naif Mohsen, Ahmed Almathloum, Ahmed Abbad, Nouradin Abdurahman	Full Stack Engineer	High	Completed	03/05/2026 → 09/05/2026	Documentation	Seddik Mohey Abdelhamid Seddik, Nouraddin Abdurahman Aden, Ahmed Adeeb Ahmed Abbad, Mohsen Naef Qahtan, Ahmed K.A. Almathloum	**Technical Documentation:** Finalizing system architecture, database schemas, and AI logic explanations. **Visual Design:** Designing the graduation poster, presentation slides, and branding assets. **Presentation Prep:** Organizing the live demo flow, feature highlights, and technical explanations. **Evidence Gathering:** Preparing GitHub evidence, project screenshots, and workflow diagrams.	**Academic Assets:** Final Project Report & Graduation Poster. **Presentation:** Final Presentation Slides & Demo-ready MVP. **Technical:** Full Architecture & Backend Documentation.
Sprint 13 — Testing & Quality Assurance	Sprint 13	Seddik Mohey, Naif Mohsen, Ahmed Almathloum, Ahmed Abbad, Nouradin Abdurahman	Full Stack Engineer	High	Completed	26/04/2026 → 02/05/2026	Testing/QA	Ahmed K.A. Almathloum, Mohsen Naef Qahtan, Ahmed Adeeb Ahmed Abbad, Nouraddin Abdurahman Aden, Seddik Mohey Abdelhamid Seddik	**Logic & AI Validation:** Rigorous testing of the Risk Engine, AI Assistant, and Analytics accuracy. **Architecture & Sync:** Stress testing realtime synchronization, state management, and integration flows. **Security & Auth:** Security validation, authentication flow testing, and backend stability. **Frontend QA:** Cross-device testing, UI responsiveness, and manual UX flow validation. **Performance:** Regression testing and bug fixing across all modules.	- **Reports:** Comprehensive QA Report and Bug Tracking Report - **Validation:** Stability Validation and documented Testing Evidence.
Sprint 12 — App Stabilization & Final Feature Completion	Sprint 12	Ahmed Almathloum	Front End Developer	Medium	Completed	19/04/2026 → 25/04/2026	Analytics	Ahmed K.A. Almathloum	Conduct final UI review & fix visual edge cases. Improve charts/graphs and general readability. Finalize all animations and interaction feedback. Ensure visual consistency across the entire application.	Finalized visual design, Polished user experience.
Sprint 12 — App Stabilization & Final Feature Completion	Sprint 12	Naif Mohsen	UI/UX Designer	High	Completed	19/04/2026 → 25/04/2026	UI/UX	Mohsen Naef Qahtan	Fix frontend visual issues & responsive design problems. Improve UX consistency & user onboarding flow. Refine animations and accessibility consistency. Validate all frontend screens for production readiness.	Stable frontend experience, Consistent mobile UI.
Sprint 12 — App Stabilization & Final Feature Completion	Sprint 12	Ahmed Abbad	Back-End Engineer	High	Completed	19/04/2026 → 25/04/2026	Authentication	Ahmed Adeeb Ahmed Abbad	Fix backend inconsistencies & optimize logging. Improve database reliability & validate realtime systems. Optimize authentication flow & validate security policies. Finalize backup and recovery handling protocols.	Stable backend infrastructure, Reliable database operations.
Sprint 12 — App Stabilization & Final Feature Completion	Sprint 12	Nouradin Abdurahman	Full Stack Engineer	High	Completed	19/04/2026 → 25/04/2026	Testing/QA	Nouraddin Abdurahman Aden	Fix integration bugs & refactor unstable providers. Improve overall app stability & architecture consistency. Fix state synchronization issues between local/remote. Validate complete application flow & finalize deployment config.	Stable application build, Optimized application flow.
Sprint 12 — App Stabilization & Final Feature Completion	Sprint 12	Seddik Mohey	Full Stack AI Engineer	High	Completed	19/04/2026 → 25/04/2026	Prediction	Seddik Mohey Abdelhamid Seddik	Fix AI edge cases & improve recommendation stability. Improve analytics consistency & validate prediction outputs. Optimize stress calculations & refactor AI-related systems. Improve fallback handling for system failures.	Stable AI assistant, Reliable analytics systems.
Sprint 11 — Realtime Synchronization & System Optimization	Sprint 11	Ahmed Almathloum	Front End Developer	Medium	Completed	12/04/2026 → 18/04/2026	Analytics	Ahmed K.A. Almathloum	Final UI polishing & component system refinement. Improve dark/light theme consistency. Enhance charts, analytics visuals, and dashboard readability. Improve interaction feedback and finalize visual assets.	Final polished UI, Consistent visual system, Improved dashboa experience.
Sprint 11 — Realtime Synchronization & System Optimization	Sprint 11	Naif Mohsen	Front End Developer	Medium	Completed	12/04/2026 → 18/04/2026	Tasks & Deadlines	Mohsen Naef Qahtan	Improve mobile responsiveness & usability. Finalize onboarding UX & navigation flow. Polish screen layouts, transitions, and animations.Improve accessibility and finalize visual consistency.	Refined UI/UX system, Improved application usability.
Sprint 11 — Realtime Synchronization & System Optimization	Sprint 11	Ahmed Abbad	Back-End Engineer	High	Completed	12/04/2026 → 18/04/2026	Architecture	Ahmed Adeeb Ahmed Abbad	Optimize Supabase queries & configure database indexes. Improve backend response performance & scalability. Optimize realtime subscriptions & aggregation queries. Harden database security & finalize migration structure.	Optimized database performance, Improved backend scalabilit Hardened backend security.
Sprint 11 — Realtime Synchronization & System Optimization	Sprint 11	Nouradin Abdurahman	Full Stack Engineer	High	Completed	12/04/2026 → 18/04/2026	Realtime & Performance	Nouraddin Abdurahman Aden	• Integrate realtime providers & optimize invalidation flows. • Finalize architecture integration & state management. • Improve app startup performance & memory handling. • Optimize provider dependencies and navigation flows.	Stable realtime synchronization, Optimized application architecture, Improved performance.
Sprint 11 — Realtime Synchronization & System Optimization	Sprint 11	Seddik Mohey	Full Stack AI Engineer	High	Completed	12/04/2026 → 18/04/2026	Prediction	Seddik Mohey Abdelhamid Seddik	Optimize analytics calculations & prediction accuracy. Improve recommendation quality & burnout detection precision. Finalize Risk Engine tuning and refactor AI-related logic. Optimize AI response handling and integration performance.	Optimized AI systems, Improved prediction quality, Finalized recommendation engine.
Sprint 10 — Smart Notification System	Sprint 10	Ahmed Almathloum	Front End Developer	Medium	Completed	05/04/2026 → 11/04/2026	Notifications	Ahmed K.A. Almathloum	Improve notification UX & interaction flow. Create alert animations & visual notification hierarchy. Improve readability, accessibility, and responsive layouts. Create notification empty states.	Notification visual system, Mobile notification UX improvement
Sprint 10 — Smart Notification System	Sprint 10	Naif Mohsen	Front End Developer	Medium	Completed	05/04/2026 → 11/04/2026	Notifications	Mohsen Naef Qahtan	• Build notification inbox UI & message cards. • Create academic alert UI & reminder components. • Build swipe notification interactions & animations. • Design notification states & improve accessibility.	Notification screens, Academic alert components, Improved mobile interactions.
Sprint 10 — Smart Notification System	Sprint 10	Ahmed Abbad	Back-End Engineer	High	Completed	05/04/2026 → 11/04/2026	Notifications	Ahmed Adeeb Ahmed Abbad	Configure Firebase Cloud Messaging (FCM) & Supabase backend. Build notification database structure & push token persistence. Configure Edge Function push delivery & realtime listeners. Build background notification worker & cooldown storage logic.	Push notification backend, Realtime architecture, Background worker system.
Sprint 10 — Smart Notification System	Sprint 10	Nouradin Abdurahman	Full Stack Engineer	High	Completed	05/04/2026 → 11/04/2026	Architecture	Nouraddin Abdurahman Aden	Integrate notification providers & local notification handling. Connect notification system with dashboard & risk engine. Build notification inbox integration & state handling. Integrate realtime updates & routing/deep linking.	Notification UI integration, Functional notification inbox, Connected realtime flow.
Sprint 10 — Smart Notification System	Sprint 10	Seddik Mohey	Full Stack AI Engineer	High	Completed	05/04/2026 → 11/04/2026	Risk Engine	Seddik Mohey Abdelhamid Seddik	• Implement SmartNotificationEngine & cooldown algorithms. • Build behavioral notification logic & risk-triggered alerts. • Implement overdue task detection & stress warning system. • Create adaptive notification prioritization & intelligent batching.	Smart notification intelligence, Adaptive notification system, Behavioral alert engine.
Sprint 9 — Prediction System & AI Assistant	Sprint 9	Ahmed Almathloum	UI/UX Designer	Medium	Completed	29/03/2026 → 04/04/2026	UI/UX	Ahmed K.A. Almathloum	Improve AI interaction UX & screen transitions Build AI recommendation cards & analytics widgets Design assistant visual states (Loading, Empty, Active) Create accessible chat experience & improve readability	AI Visual System, AI Recommendation Components, Improved Assistant UX.
Sprint 9 — Prediction System & AI Assistant	Sprint 9	Naif Mohsen	Front End Developer	Medium	Completed	29/03/2026 → 04/04/2026	UI/UX	Mohsen Naef Qahtan	Build AI chat interface & conversation screens Create typing animations & message card UI Develop AI quick-action buttons & smooth chat interactions Improve mobile chat responsiveness	AI Chat UI, Assistant Interaction System, Conversation Experience Design.
Sprint 9 — Prediction System & AI Assistant	Sprint 9	Ahmed Abbad	Back-End Engineer	High	Completed	29/03/2026 → 04/04/2026	Architecture	Ahmed Adeeb Ahmed Abbad	Configure Supabase Edge Functions & NVIDIA NIM API Setup secure AI backend communication & API env variables Optimize AI request handling & backend response flow Build AI logging structure & rate limiting	AI Backend Infrastructure, Edge Function Deployment, Secure AI API Architecture.
Sprint 9 — Prediction System & AI Assistant	Sprint 9	Seddik Mohey	Full Stack AI Engineer	High	Completed	29/03/2026 → 04/04/2026	Prediction	Seddik Mohey Abdelhamid Seddik	Implement AiIntentParser & AiDataResolver Build academic intent recognition system Develop prediction engine calculations (performance & stress trends)	Functional AI Assistant Logic, Prediction Engine, Academic Forecasting System.

System Architecture

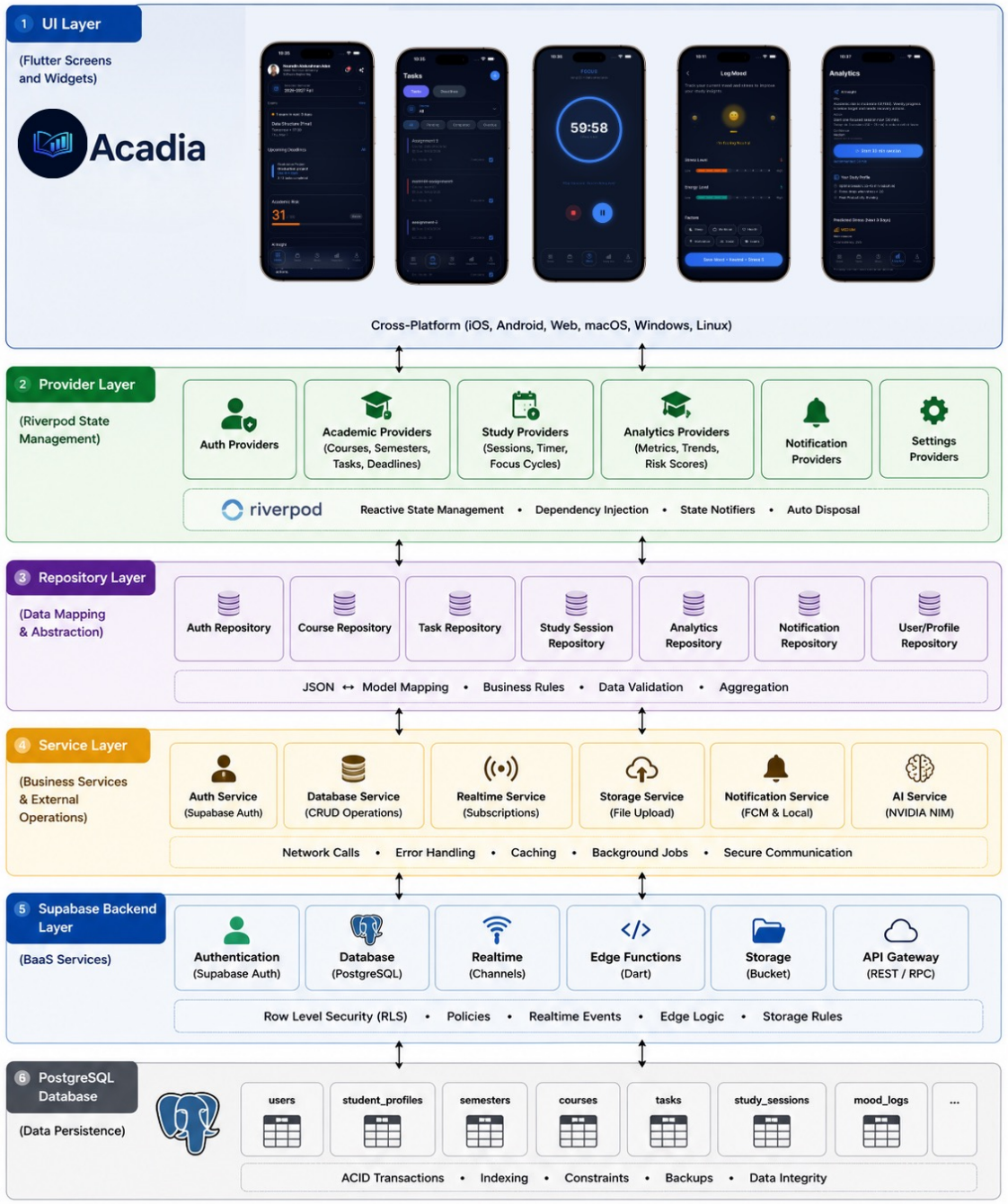
Acadia System Architecture

Acadia follows a layered architecture.
The architecture separates:

- Ui
- Providers
- Repositories
- Services
- Backend infrastructure

This improves:

- Scalability
- Maintainability
- Modularity
- Realtime synchronization



Database Design

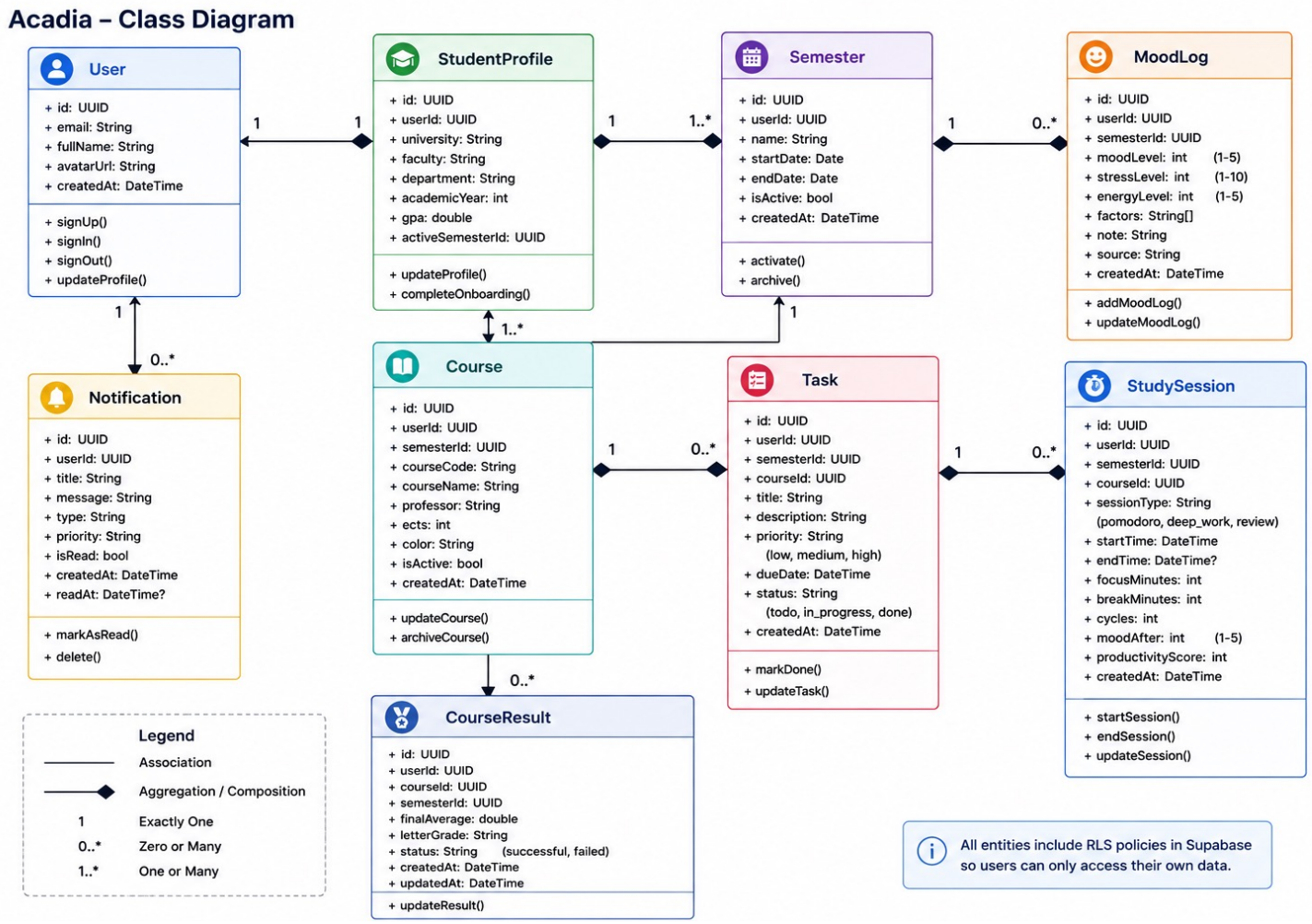
Database Structure

Acadia uses PostgreSQL through Supabase.

The database stores:

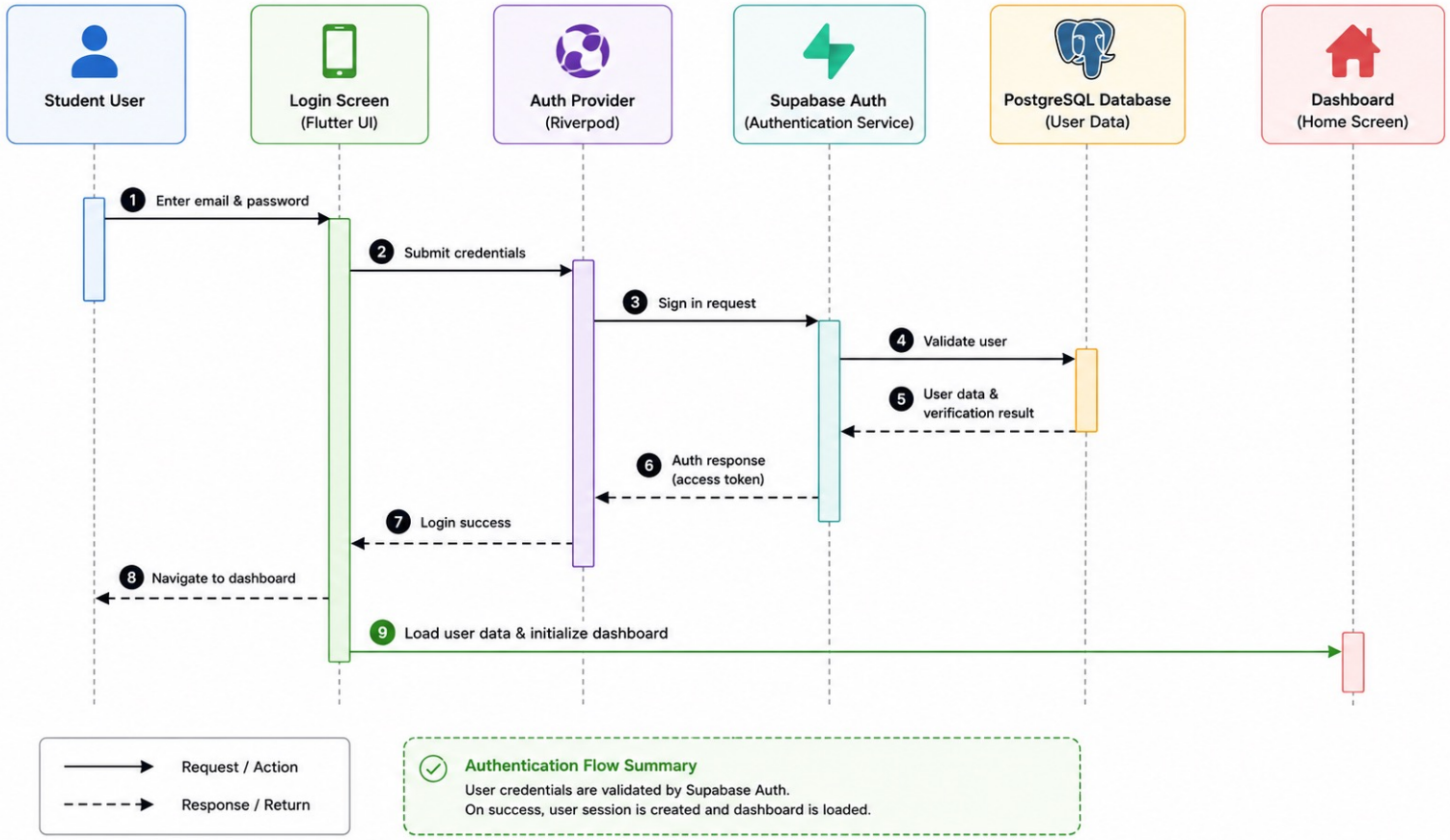
- Semesters
- Courses
- Tasks
- Exams
- Study sessions
- Analytics
- Notifications
- Mood logs

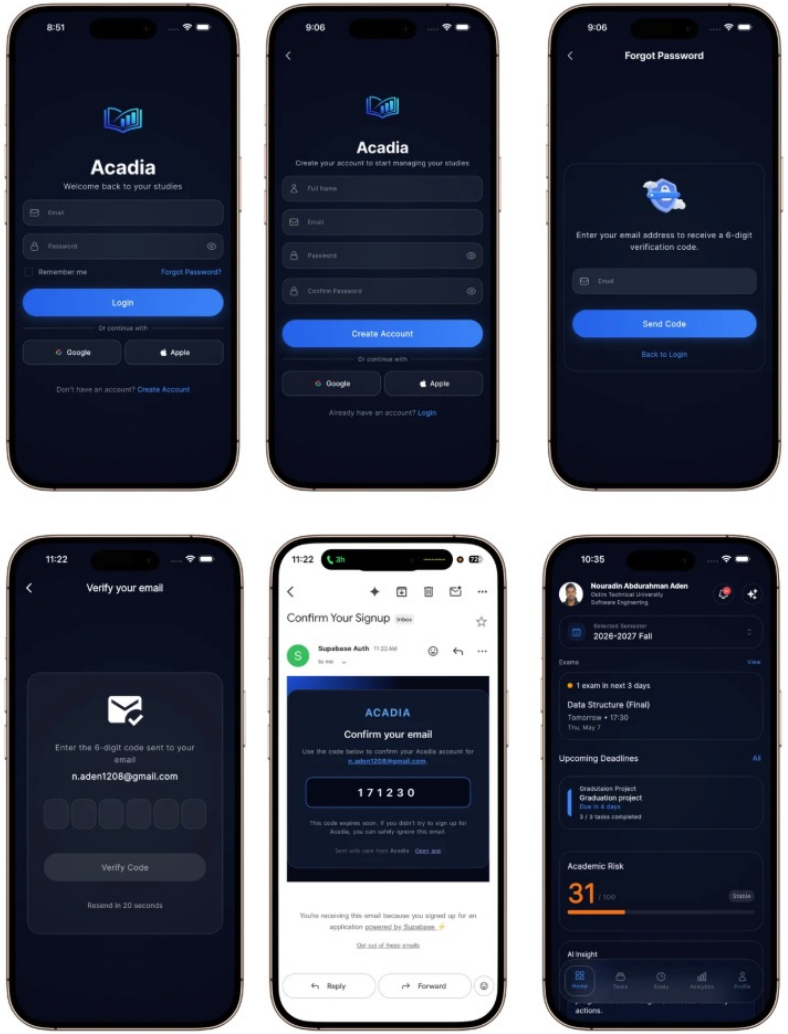
Security is implemented using Row Level Security (RLS).



Authentication System

Acadia provides secure authentication using Supabase Auth.





Academic Management System

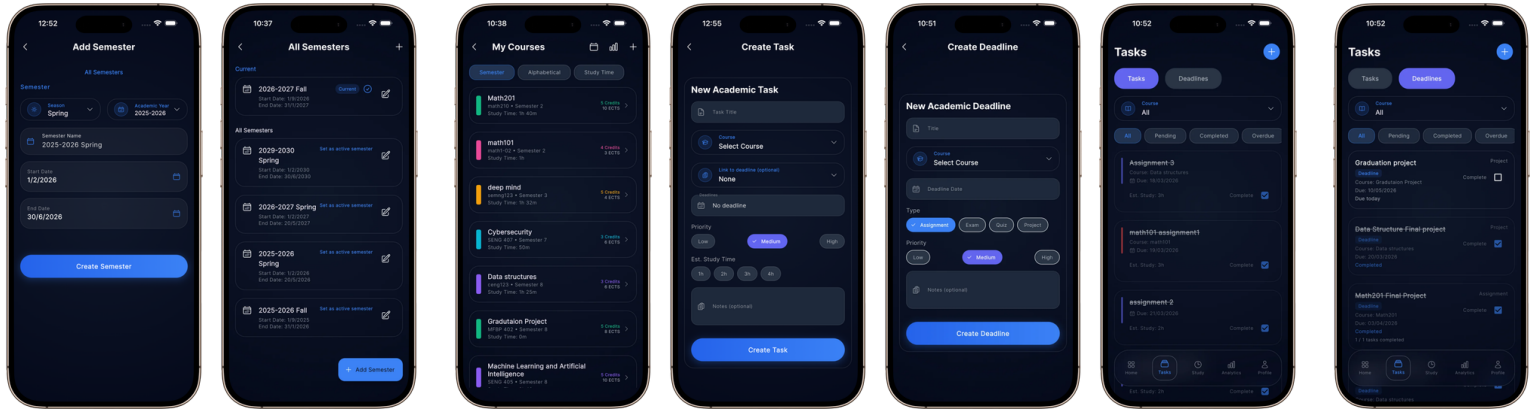
Semester, Course & Task Management

Acadia allows students to organize academic life inside one centralized system.

Students can:

1. Create semesters
2. Add courses
3. Manage assignments
4. Manage deadlines
5. Manage exams
6. Track academic progress

The system supports realtime synchronization.

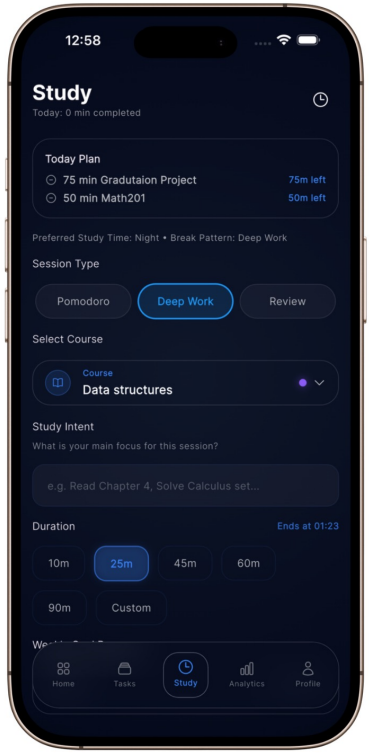


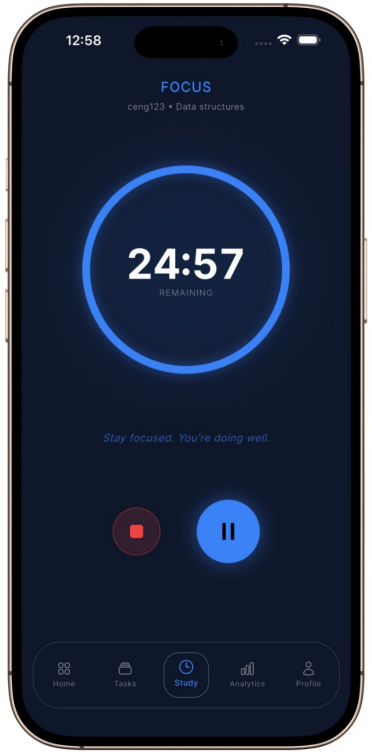
Study Session System

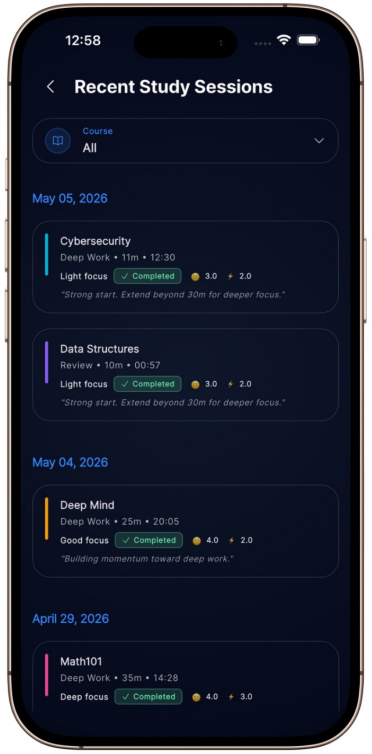
Pomodoro & Study Session System

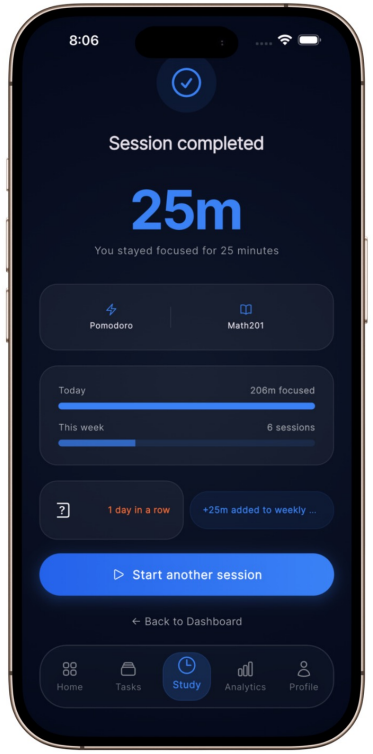
Acadia includes advanced study session tracking.
Supported modes:

1. Pomodoro sessions
2. Deep-work sessions
3. Review sessions
4. Features:
5. Focus/break cycles
6. Timer persistence
7. Session recovery
8. Productivity tracking
9. Realtime analytics updates









Academic Analytics Engine

Academic Analytics System

Acadia calculates academic metrics in realtime.
The analytics engine evaluates:

- GPA trends
- Workload pressure
- Study consistency
- Task completion rate
- Stress trends
- Burnout indicators



GPA Formula

$$GPA = \frac{\sum(Course\ Grade \times ECTS)}{\sum ECTS}$$

Productivity Formula

$$Productivity\ Score = 0.4(Task\ Completion) + 0.3(Study\ Consistency) + 0.3(Focus\ Sessions)$$

Burnout Formula

$$Burnout\ Score = Stress\ Level + Overdue\ Tasks + Study\ Deficit$$

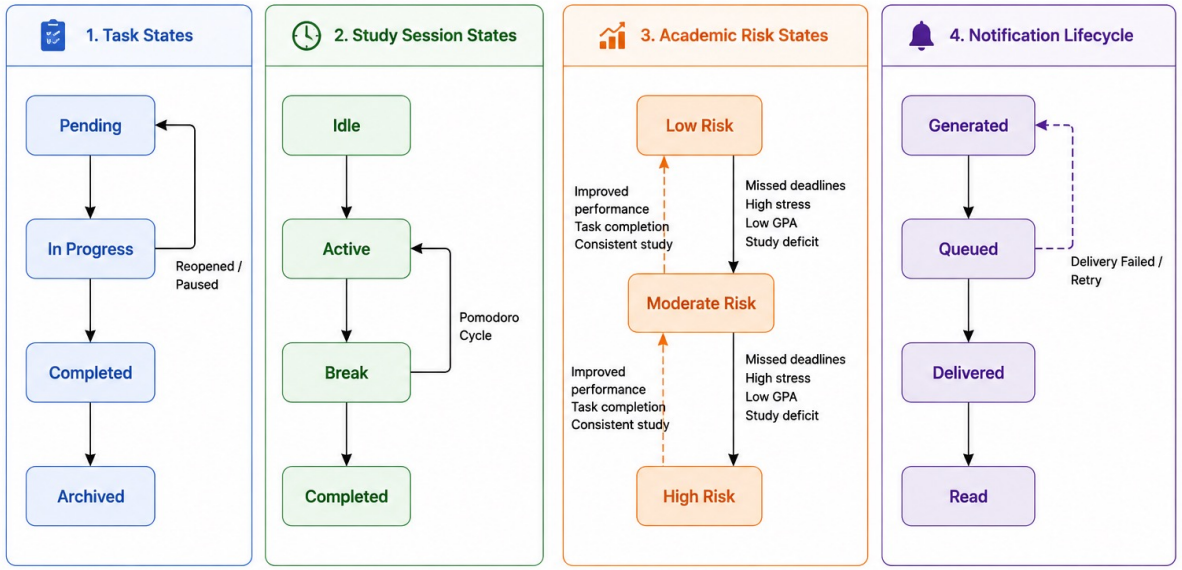
Academic Risk Engine

Academic Risk Detection System

Acadia evaluates academic risk levels using behavioral and productivity indicators.

The system analyzes:

1. Overdue tasks
2. GPA decline
3. Stress escalation
4. Workload pressure
5. Low study consistency
6. Missed deadlines



The state diagram represents the behavioral transitions within the Acadia system. Tasks, study sessions, academic risk levels, and notifications move through predefined states based on user activity, analytics calculations, and realtime system events. The model improves workflow consistency, monitoring accuracy, and intelligent notification management.

Risk Formula

$$Risk\ Score = 0.34(Urgency) + 0.24(Study\ Deficit) + 0.16(Workload) + 0.16(ECTS) + 0.10(Risk\ Signals)$$

Score	Risk Level
0–30	Low Risk
31–60	Moderate Risk
61–100	High Risk



AI Assistant

AI Assistant & Recommendation System

Acadia integrates an AI assistant for academic support.
The AI system uses:

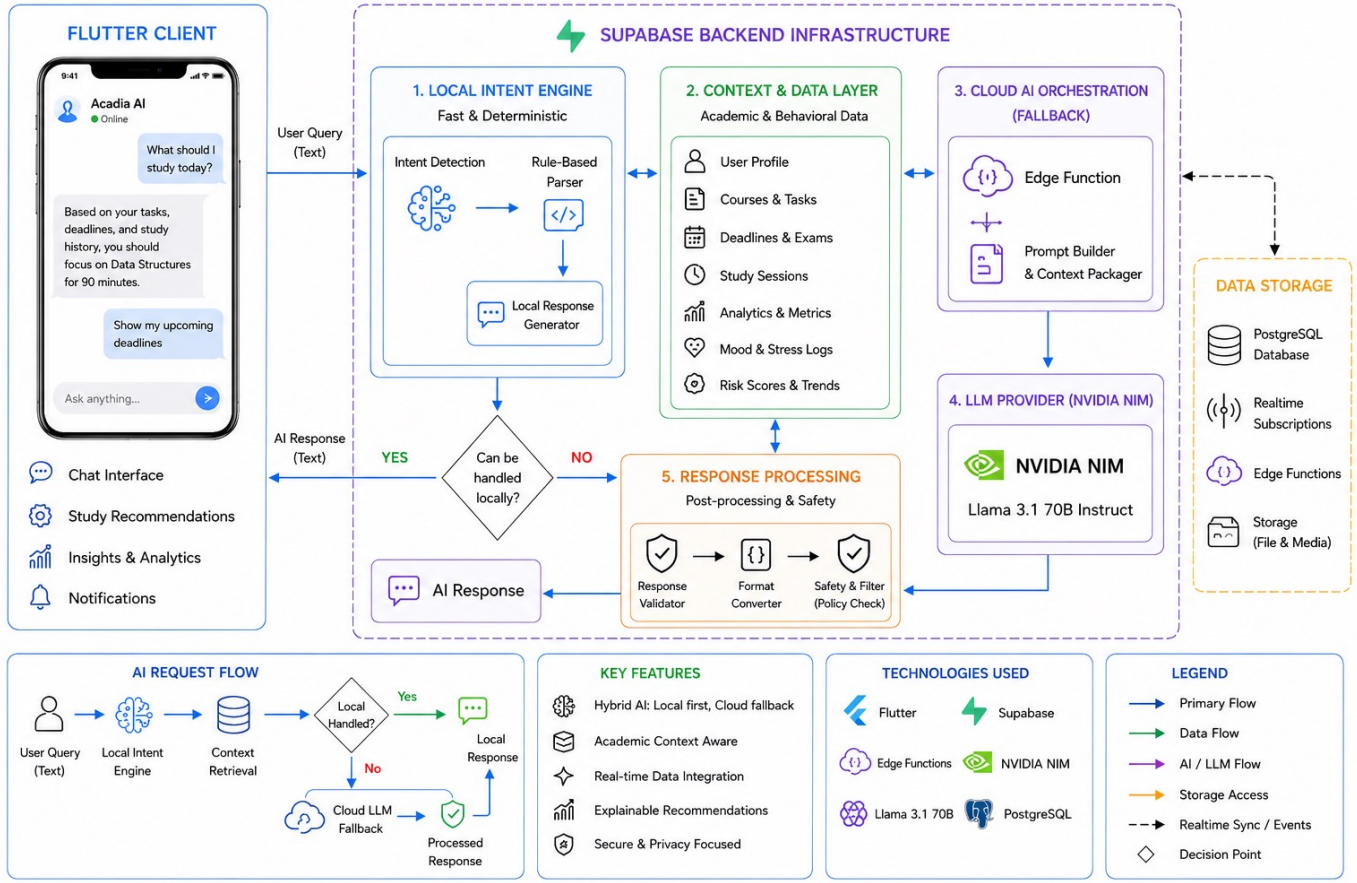
- Local intent parsing
- Recommendation logic
- Cloud AI fallback using NVIDIA NIM and llama 3.1

The assistant helps students:

- Prioritize courses
- Identify weak subjects
- Improve study planning
- Understand productivity trends

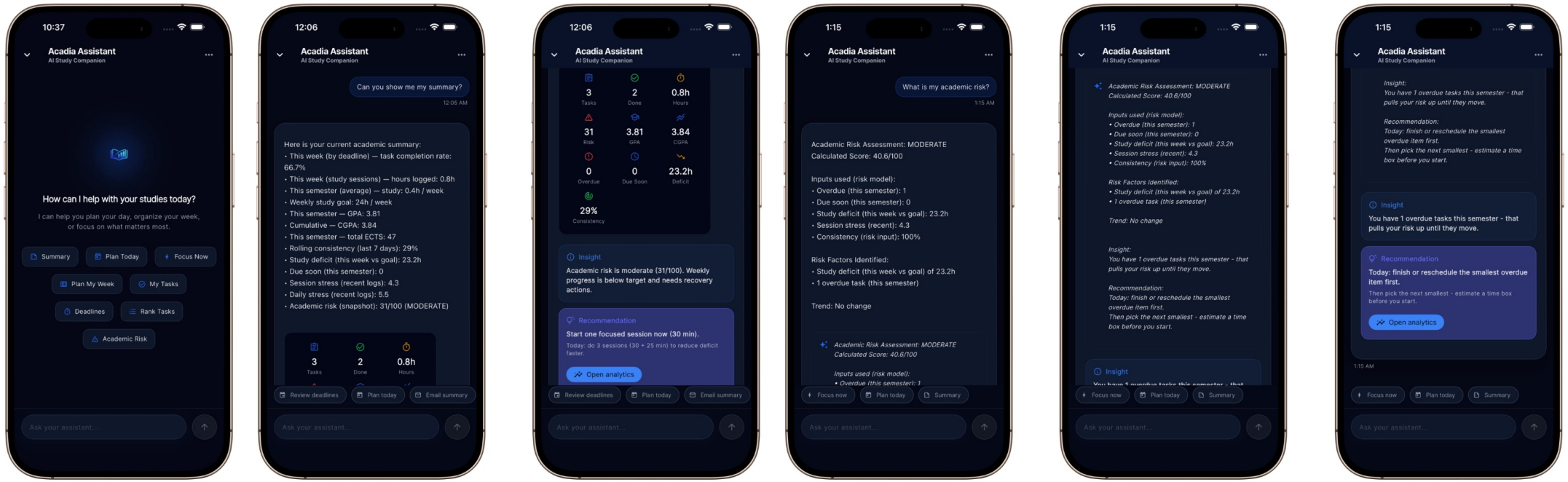
AI Workflow

User Query
→ Local Intent Engine
→ Analytics Evaluation
→ AI Recommendation Engine
→ Llama 3.1 Fallback
→ Response Generation



AI Assistant

AI Assistant & Recommendation System



Realtime Notification System

Smart Notification System

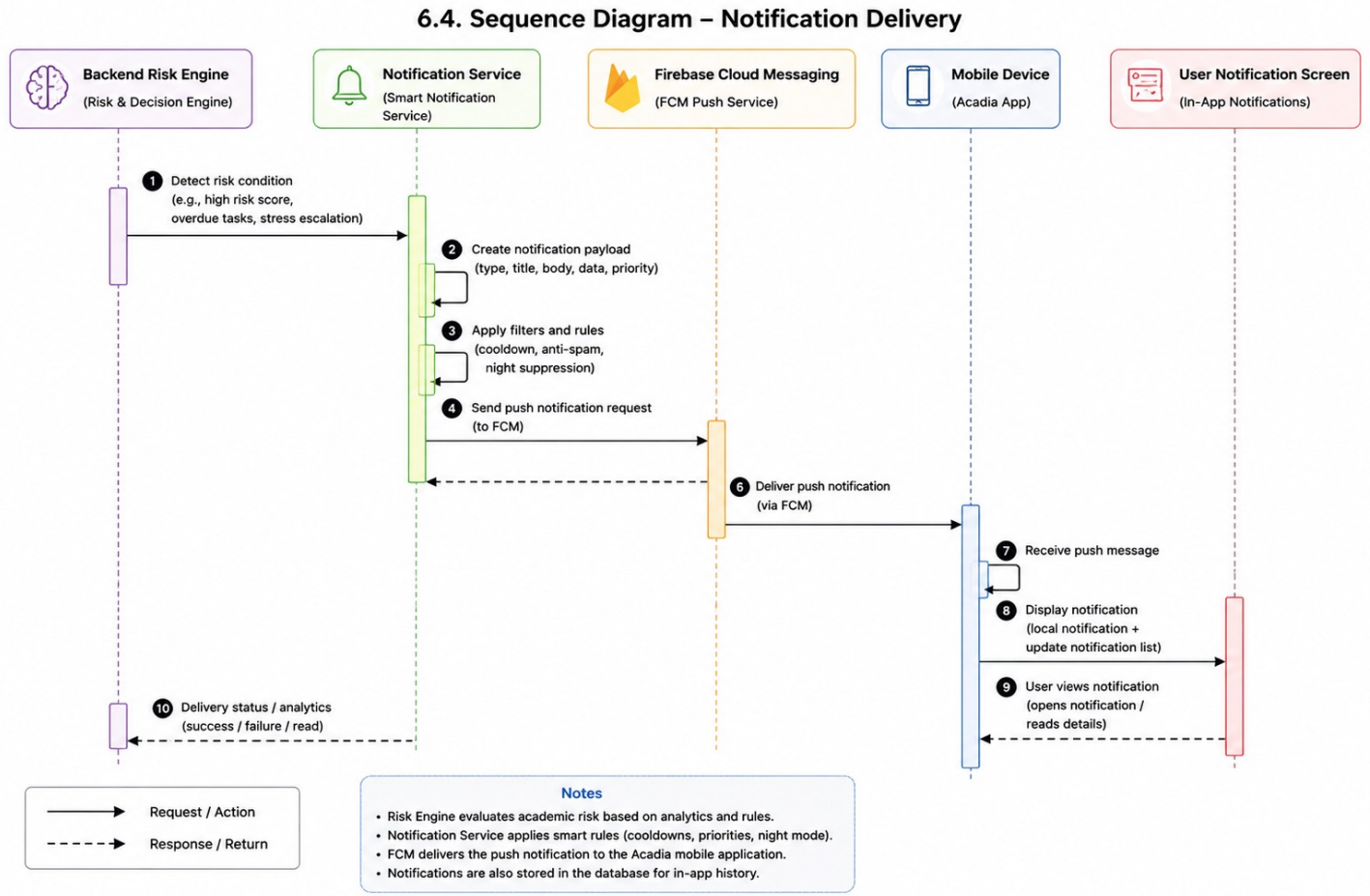
Acadia includes intelligent realtime notifications.
Notifications are generated for:

1. Overdue tasks
2. Deadline reminders
3. High stress levels
4. Academic risk alerts
5. Study recommendations

Firebase Cloud Messaging is used for push notifications

Notification Flow

Risk Engine
→ Notification Service
→ Firebase Cloud Messaging
→ Mobile Device
→ User Notification



Realtime Notification System



Testing & Evaluation

The system was tested using:

- Unit testing
- Integration testing
- Realtime synchronization testing
- Usability testing
- Scenario-based testing

Test Scenario	Result
Authentication Flow	Passed
Realtime Synchronization	Passed
Study Session Recovery	Passed
Notification Delivery	Passed
Risk Engine Calculations	Passed
Analytics Dashboard	Passed
AI Assistant Responses	Passed

Testing & Evaluation

```
A Dart VM Service on iPhone 17 Pro Max is available at: http://127.0.0.1:55493/hH_9737sjoU=/
The Flutter DevTools debugger and profiler on iPhone 17 Pro Max is available at:
http://127.0.0.1:55493/hH_9737sjoU=/devtools/?uri=ws://127.0.0.1:55493/hH_9737sjoU=/ws
flutter: [Notification] iOS local permission granted: true
flutter: [Notification] Local timezone set: Europe/Istanbul
flutter: LOGIN START
flutter: EVENT: AuthChangeEvent.signedIn
flutter: SESSION: not null
flutter: SIGNED IN SUCCESSFULLY
flutter: [ensurePublicUser] called — user: 0ee42413-7a9c-4740-8b6f-97667ba1b913
flutter: [extractName] metadata keys: [email, email_verified, full_name, phone_verified, sub]
flutter: [ensurePublicUser] extracted name: "Nouradin Abdurahman"
flutter: Fetching current user from database...
flutter: Auth user id: 0ee42413-7a9c-4740-8b6f-97667ba1b913
flutter: FCM TOKEN: f2_iJd3jc0wonxeQkbj2Yl:APA91bHmux_zBfE0Q7FGf0ZbtXY1CDboWjtpzP_A_DriEGmxxrKtkv70i6030aheEqjgW_0kb_1AHOdWrt_Qz757lg0vqmrVZZmjThDE3i76j10PWNHyUpo
flutter: LOGIN SUCCESS
flutter: Fetched user: Instance of 'AppUser'
```

LOG IN PROCESS

```
flutter: [ActiveSemester] build() → initial load from DB: 6825dfc9-98b9-4d28-a280-911fb9418cf3
flutter: [AnalyticsProvider] Found active semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3. Starting data fetch...
flutter: [TasksProvider] Loading tasks for user: 0ee42413-7a9c-4740-8b6f-97667ba1b913, semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3
flutter: [TasksNotifier] Loading tasks for user: 0ee42413-7a9c-4740-8b6f-97667ba1b913, semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3
flutter: [DeadlinesProvider] Loading deadlines for user: 0ee42413-7a9c-4740-8b6f-97667ba1b913, semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3
flutter: [CourseProvider] Loading courses for semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3
flutter: [AnalyticsProvider] Found active semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3. Starting data fetch...
flutter: [ExamsProvider] Loading exams for user: 0ee42413-7a9c-4740-8b6f-97667ba1b913, semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3
flutter: [ensurePublicUser] users row ready ✓
flutter: [ensurePublicUser] student_profiles row ready ✓
flutter: [StudySessionsProvider] Loading for Semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3 | Data Count: 22
flutter: Fetching current user from database...
flutter: Auth user id: 0ee42413-7a9c-4740-8b6f-97667ba1b913
flutter: [AnalyticsProvider] Found active semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3. Starting data fetch...
flutter: [AnalyticsProvider] Found active semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3. Starting data fetch...
flutter: [TasksNotifier] Load complete. Semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3 | Data Count: 19
flutter: [AnalyticsProvider] Found active semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3. Starting data fetch...
flutter: Fetched user: Instance of 'AppUser'
flutter: [CourseProvider] Loading for Semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3 | Data Count: 7
flutter: [MoodProvider] Loading for Semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3 | Data Count: 24
flutter: [AnalyticsProvider] Found active semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3. Starting data fetch...
flutter: [AnalyticsProvider] Computing for Semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3 | Sessions: 22 | Tasks: 19 | Moods: 24 | Deadlines: 6
flutter: [Exams] Loaded 4 exams for semester 6825dfc9-98b9-4d28-a280-911fb9418cf3
flutter: [AnalyticsRepository] computeFromPreloaded Detail:
flutter:   - Semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3
flutter:   - Total Moods: 24
flutter:   - Daily Classified: 17
flutter:   - Session Classified: 7
flutter: [AnalyticsProvider] Computed for Semester: 6825dfc9-98b9-4d28-a280-911fb9418cf3 | Sessions: 22 | Tasks: 19 | Moods: 24 | Deadlines: 6
flutter: [AI Insight] hasData=true hasSignal=true
flutter: [AI Insight] hasData=true hasSignal=true
flutter: [AI Insight] hasData=true hasSignal=true
flutter: [AI Insight] hasData=true hasSignal=true
```

USER DATA FETCHING

Deployment & Publishing

Acadia was deployed as a production-ready application.
The project includes:

1. App store publishing
2. Production supabase backend
3. Realtime cloud infrastructure
4. Mobile deployment pipeline
5. Github repository

A dedicated jury testing account was created.



Deployment & Publishing

Jury Account

Email: jury@acadia.app

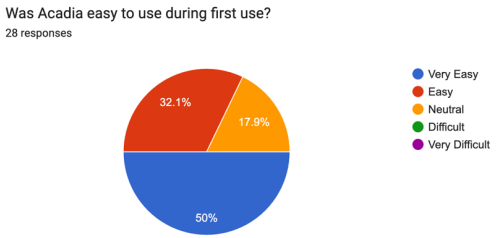
Password: Acadia2026

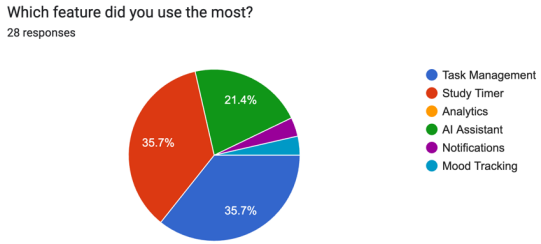
User Feedback & Survey

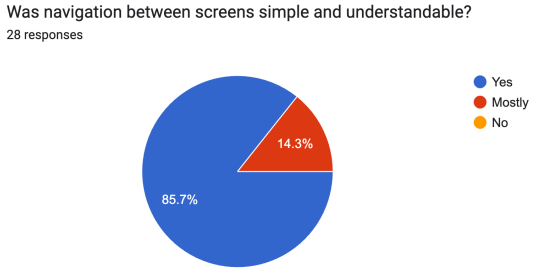
User Feedback and Evaluation

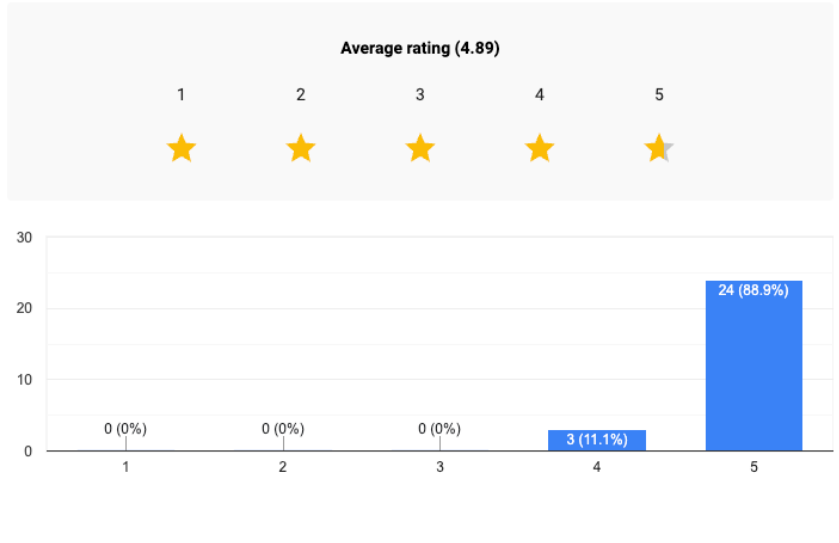
To evaluate Acadia in a real-world environment, the application was deployed on the Apple App Store and tested by university students.

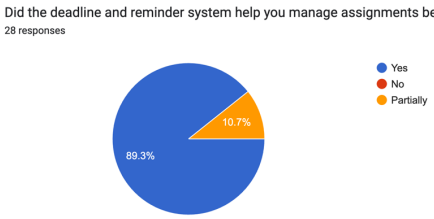
A survey was conducted with 28 participants to measure usability, usefulness, UI/UX quality, feature effectiveness, AI recommendation quality, and system stability.

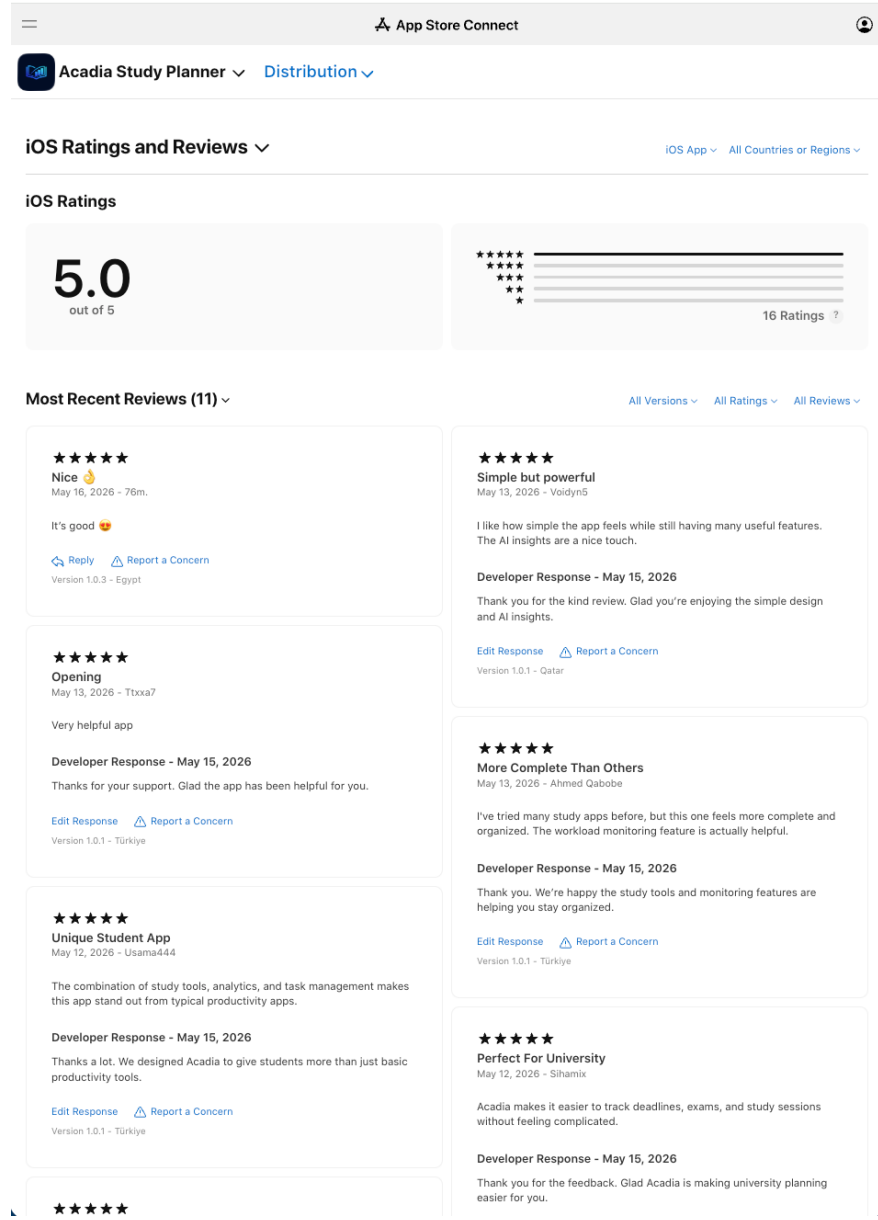












Challenges Faced

Challenges During Development

Main challenges included:

1. Realtime synchronization complexity
2. Provider state consistency
3. Notification scheduling
4. Timer persistence
5. AI orchestration
6. Supabase realtime handling
7. Analytics optimization
8. Cross-platform UI consistency

The team solved these challenges through iterative Scrum development and continuous testing.

Future Work

Future Improvements

Future development plans include:

1. Predictive academic forecasting
2. Advanced AI tutoring
3. Calendar synchronization
4. Wearable device integration
5. Collaborative study groups
6. Advanced machine learning models
7. University system integration
8. Web dashboard expansion

Conclusion

Acadia demonstrates how software engineering, AI, analytics, and realtime systems can be combined into one academic productivity ecosystem.

Acadia provides:

1. Intelligent academic monitoring
2. Productivity analytics
3. Realtime academic insights
4. Ai-assisted recommendations
5. Stress and burnout awareness

The project achieved its objectives and produced a scalable production-ready MVP.



App Store

GitHub

Notion

Survay

OSTİM TEKNİK ÜNİVERSİTESİ
ANKARA

Thank You

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